When does a corpus-known opposition stay binary in residual space?

A process-oriented dichotomy probe using residual representations as the unit of analysis

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Dichotomy Transformation Probe v2 - revised publishable draft

Research question: Given a corpus-known named dichotomy, does the transformer's evolving residual representation preserve the relation as an exhausted two-pole field, or does layerwise transformation produce stable middle/off-axis structure despite the binary frame?

Short answer: The central result is a dissociation: pole co-situation does not imply middle seating. In heads/tails, the poles become highly co-situated across depth while every candidate middle remains off-axis or prompt-induced. Scalar pairs such as hot/cold and wet/dry behave differently, producing stable midpoint candidates. The good/bad phrase-level result is promising but provisional pending phrase-representation and lexical-echo controls.

Design rationale: The probe intentionally uses corpus-known, culturally stabilized oppositions. The goal is not to discover whether the model has encountered these relations, but to observe what the residual stream does with already-stabilized relational forms across depth.

Unit-of-analysis guardrail: Vygotsky's X-unit, word meaning, motivates the methodological question but is not attributed to the transformer. The measured Y-unit is the evolving residual representation.

Repository: https://github.com/jenniferchamberspalmer-research/token-biography

Figure source branch: jenniferchamberspalmer-research/What-if-Conceptual-Turn-in-MI-, branch dichotomy-probe, folder outputs/dichotomy_probe_v2/figures/

Model: Gemma 2 2B base. Residual reads: 27 states across layers 0 to output. Unit of analysis: residual representation at a fixed token position.

Status: Exploratory mechanistic interpretability report. v2 is publishable as an exploratory result; v3 controls are recommended for the phrase-level good/bad claim and cosine baseline.

1. Why this probe

This project starts from a unit-of-analysis problem. Transformers are often described by their final behavior: they predict the next token. But prediction is the final readout of a sequence of internal transformations. If we want to study the process by which a transformer turns a corpus-shaped token state into a context-specific computational state, then the token ID, final output, attention head, or MLP block alone is not the right unit.

The proposed Y-unit is the evolving residual representation: a corpus-shaped token-position state transformed across depth by learned transformer operations, including attention-mediated contextual integration and MLP-mediated internal transformation.

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Y-unit = evolving residual representation

evolving residual representation =
corpus-shaped token-position state
+
learned transformer operations
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This probe tests that unit in one constrained domain: corpus-known named dichotomies. The goal is not to discover whether the corpus contains heads/tails, even/odd, or hot/cold relations. Those relations are already culturally and linguistically stabilized. The goal is to ask what the transformer's residual stream does with a stabilized relation once it is instantiated in a prompt.

In this sense, the dichotomy probe functions as a controlled relation chamber. By selecting corpus-known named oppositions, it reduces lexical ambiguity enough to observe how the residual stream instantiates relational structure across depth. The central finding is that the residual stream can make a relation geometrically legible without collapsing it into a scalar continuum.

The core distinction is between relation-field visibility and middle/value-field exhaustion. A pair such as heads/tails may become geometrically close because the model processes the shared coin-flip relation. But that does not mean the field contains a middle. A pair such as hot/cold may also become geometrically close, but if terms like warm, mild, and lukewarm stably seat between the poles, then the residual geometry suggests a scalar relation-field rather than an exhausted binary.

The paper does not ask whether the model knows what a true dichotomy is. It asks whether, given a corpus-known named opposition, the residual representation preserves a two-pole field, produces stable middle structure, or generates off-axis/span-dependent structure.

2. Unit-of-analysis note: why X and Y are kept separate

This project is motivated by a unit-of-analysis question, but it does not collapse human meaning into transformer internals.

In Vygotsky's framework, the X-unit is word meaning: the functional whole in which socially mediated speech and reorganized psychological activity are united in consciousness. That unit belongs to human development and should not be attributed to the transformer.

The empirical object in this paper is Y: the evolving residual representation. In Y, the residual representation is a corpus-shaped token-position state transformed across depth by learned transformer operations. Transformer readouts are not treated as sites where human meaning is made. They are treated as observable computational traces of how a trained language system transforms corpus-shaped relations.

The analogy is methodological, not ontological: X motivates the question of what kind of unit can preserve a dynamic process, while Y names the computational unit actually measured in this probe.

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X-unit: word meaning = socially mediated speech + reorganized psychological activity

Y-unit: evolving residual representation = corpus-shaped token-position state + learned
transformer operations
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The study does not claim that residual representations are word meanings; it uses residual representations as measurable traces of relation-structure transformation inside a trained transformer.

3. Related work and methodological perimeter

This probe sits at the intersection of residual-stream mechanistic interpretability, contextual representation geometry, layerwise prediction formation, and relation decoding.

Residual-stream work in transformer circuits establishes the residual stream as the central read/write channel of transformer computation. The present study builds on that view but shifts the question from component analysis alone to process-bearing representation: what does the evolving residual representation hold across depth?

A first methodological constraint is anisotropy. Contextualized representations are not isotropic at any layer, so raw cosine similarity cannot by itself establish relation-specific pole co-situation. For that reason, v3 should report cosine_AB_excess: pole cosine minus unrelated-pair cosine under the same carrier and layer. This would make pole co-situation structurally parallel to remainder_vs_control, where candidate seating is interpreted relative to an unrelated-word baseline.

A second constraint is output-facing compression. Prior work on layerwise representation evolution and prediction formation suggests that upper layers increasingly construct and select predictions. In this context, the late-layer good/bad pattern is best interpreted descriptively: relation proximity can remain geometrically legible across most of depth and decohere only near the output-facing stage.

The current paper does not explain why that happens; it establishes the empirical separation between relation holding and final predictive commitment.

The probe also sits beside work treating relations as computational objects in transformer models. Relation-decoding work asks whether a relation can be modeled as a function from one representation to another. This probe asks a different but adjacent question: whether a named opposition forms a residual-geometric field with measurable pole, middle, and off-axis structure. The shared premise is that relations are not reducible to isolated token identities.

4. Experimental design

4.1 Model and residual reads

The probe used Gemma 2 2B base and read 27 residual states, from layer 0 to output. Tokenization fixes the position, but the token itself is not the unit. The unit is the residual representation carried at that token position and read across layers.

4.2 Pairs

Pair	Human experimental label
heads / tails	true dichotomy
even / odd	true dichotomy
hot / cold	scalar collapse
wet / dry	scalar collapse
good / bad	scalar collapse / morally complex opposition

These labels are human design labels. They are not machine verdicts. The machine never determines true or false; it transforms a cultural dichotomy already named in language, and the verdict lives in the human reader.

4.3 Carrier conditions

Carrier	Template
matched syntax	The [frame_noun] was [word].
natural usage	The coin came up heads. / The integer is even. / The temperature was hot.
explicit dichotomy	The dichotomy between [A] and [B] is familiar.
neutral relation	The relation between [A] and [B] is familiar.

The goal was to separate the dichotomy frame from the carrier frame. Carrier frames are not neutral. They are part of the input that produces the residual state.

4.4 Candidate middles

Candidate middles were tested for every pair, including true dichotomies. This was essential because 'no middle' should be a result, not an assumption.

Pair	Frame term	Candidate middles
heads/tails	coin	edge, side, rim, face, both, neither

even/odd	integer	zero, half, fraction, decimal, both, neither
hot/cold	temperature	warm, cool, mild, lukewarm, both, neither
wet/dry	moisture	damp, moist, humid, soaked, both, neither
good/bad	morality	neutral, mixed, ambiguous, gray, morally ambiguous, neither good nor bad, both good and bad

4.5 Hidden-middle forcing prompts

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Between [A] and [B], the middle case is [M].
An ambiguous case between [A] and [B] is [M].
Something neither [A] nor [B] is [M].
Something both [A] and [B] is [M].
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These prompts tested whether a candidate was stable under ordinary carriers or only became midpoint-like when the prompt forced a middle.

5. Metrics

Metric	Meaning
cosine_AB	how the two poles co-situate under each carrier
midpoint_t	candidate projection on the A to B axis; 0 = A, 1 = B, about 0.5 = centered
remainder_ratio	candidate distance off the pole axis divided by pole gap
remainder_vs_control	candidate off-axis remainder divided by unrelated-word baseline
carrier_stability	agreement of midpoint_t across word-slot carriers; only interpretable when carriers are textually distinct
mean_t_forced	candidate's mean projection under four hidden-middle forcing prompts

Absolute off-axis magnitudes are not interpretable by themselves in a 2304-dimensional residual space, so candidate seating is judged relative to the unrelated-word control baseline.

The v2 true midpoint heuristic is: 0.3 <= midpoint_t <= 0.7, remainder_vs_control <= 0.65, and carrier_stability >= 0.7. These descriptive classes are reader aids, not verdicts. The raw per-layer measurements remain the record.

Off-axis structure is also a diagnostic, not proof of a third encoding. A candidate is not interpreted as stable structure merely because it has high off-axis remainder. In v2, off-axis remainder means the candidate is not well explained by the A to B pole axis. Distinguishing structured third-position encoding from high-dimensional noise requires unrelated-pair baselines, carrier variation, phrase-representation audits, and eventually causal intervention.

6. Results

6.1 Pole co-situation across depth

Across pairs, poles generally became more co-situated across depth under matched or natural carriers. For heads/tails, output cosine reached 0.9289 under matched syntax and 0.9310 under natural usage. For even/odd, output cosine reached 0.8821 under matched syntax and 0.9662 under natural usage.

The explicit-dichotomy carrier was often lower than matched or natural usage. For example, heads/tails under explicit dichotomy reached 0.5682 at output, and even/odd reached 0.5951. Naming the relation abstractly did not strengthen pole co-situation; in these cases it weakened it relative to natural use.

[Figure 1A figure file to attach: outputs/dichotomy_probe_v2/figures/fig1a_pole_cosine_matched_syntax.svg]

Figure 1A. Pole co-situation across depth under matched-syntax carrier. cosine(A,B) by layer for all five pairs under the matched-syntax carrier. One line per pair; solid lines indicate true dichotomies and dashed lines indicate scalar-collapse pairs.

[Figure 1B figure file to attach: outputs/dichotomy_probe_v2/figures/fig1b_pole_cosine_explicit_dichotomy.svg]

Figure 1B. Pole co-situation across depth under explicit-dichotomy carrier. cosine(A,B) by layer for all five pairs under the explicit-dichotomy carrier. One line per pair; solid lines indicate true dichotomies and dashed lines indicate scalar-collapse pairs.

6.2 True dichotomies showed pole co-situation without stable midpoint candidates

This is the load-bearing result. In heads/tails, the poles reach approximately 0.93 cosine under matched/natural carriers, yet every candidate middle remains off-axis or prompt-induced. Relation-field visibility and middle seating dissociate.

heads/tails candidate	mean t	rem/ctrl	stability	class
edge	0.548	0.939	0.923	prompt-induced
side	0.501	0.929	0.926	prompt-induced
rim	0.606	0.987	0.900	prompt-induced
face	0.448	0.855	0.919	prompt-induced
both	0.524	0.990	0.900	prompt-induced
neither	0.531	1.009	0.902	prompt-induced

even/odd candidate	mean t	rem/ctrl	stability	class
zero	0.451	0.820	0.876	prompt-induced
half	0.327	0.934	0.891	prompt-induced
fraction	0.456	0.988	0.885	prompt-induced
decimal	0.533	0.969	0.893	prompt-induced
both	0.256	0.921	0.845	prompt-induced
neither	0.299	0.955	0.856	prompt-induced

These candidates often project somewhere near the middle of the A to B axis, but their off-axis remainder remains too high relative to controls. They do not count as stable midpoint candidates.

6.3 Scalar pairs produced stable midpoint candidates

Scalar pairs behaved differently. For hot/cold, warm, mild, and lukewarm met the true midpoint heuristic. For wet/dry, damp, moist, and humid met the true midpoint heuristic.

Pair	candidate	mean t	rem/ctrl	stability	class
hot/cold	warm	0.392	0.460	1.000	true midpoint
hot/cold	mild	0.559	0.623	1.000	true midpoint
hot/cold	lukewarm	0.523	0.637	1.000	true midpoint
wet/dry	damp	0.317	0.569	1.000	true midpoint
wet/dry	moist	0.380	0.534	1.000	true midpoint
wet/dry	humid	0.398	0.642	1.000	true midpoint

The 1.000 carrier-stability values should be read cautiously where matched and natural carriers are textually identical. In those cases, stability is not substantive evidence; it is a sign that v3 should mark carrier stability as trivial_identical_carrier or not applicable.

[Figure 2 figure file to attach: outputs/dichotomy_probe_v2/figures/fig2_candidate_middle_seating.svg]

Figure 2. Middle seating separates scalar fields from true dichotomies. Scatter of every candidate middle, including single-token candidates, logical terms, and phrase middles; frame terms and unrelated controls are excluded. The x-axis is mean midpoint_t, where 0 = pole A and 1 = pole B. The y-axis is remainder_vs_control. Colour indicates descriptive class; marker indicates pair. Reference lines are shown at t = 0.3, t = 0.7, and y = 0.65. Candidates in the lower-middle band sit on the pole axis more tightly than an unrelated word; the true-dichotomy candidates do not.

6.4 Good/bad shows that span-level analysis is necessary, but the phrase finding is provisional

The good/bad case should be stated cautiously. Single-token candidates did not meet the true midpoint heuristic. Phrase-level candidates produced promising midpoint behavior, but the passing phrases contain both pole tokens. Therefore, the current result shows that span-level analysis is necessary, not yet that moral middles were cleanly found.

Candidate	mean t	rem/ctrl	class
neutral	0.619	0.874	prompt-induced
mixed	0.551	0.894	prompt-induced
ambiguous	0.522	0.843	prompt-induced
gray	0.695	0.908	prompt-induced
morally ambiguous	0.530	0.874	frame-adjacent
neither good nor bad	0.552	0.630	true midpoint - provisional
both good and bad	0.477	0.575	true midpoint - provisional

Morally ambiguous is informative because it contains neither pole and lands frame-adjacent rather than midpoint-like. This suggests that phrase form and pole-token inclusion matter. v3 should add lexical-echo controls such as more good than bad, more bad than good, good versus bad, and non-pole middle phrases such as ethically ambiguous and morally neutral.

[Figure 3 figure file to attach: outputs/dichotomy_probe_v2/figures/fig3_good_bad_phrase_middles.svg]

Figure 3. Good/bad middle structure appears at phrase level. Good/bad candidates only. The x-axis is mean midpoint_t, where 0 = good and 1 = bad. The y-axis is remainder_vs_control. Stars indicate phrase middles; circles indicate single-token candidates. The phrases both good and bad (0.57) and neither good nor bad (0.63) fall below 0.65; the single-token candidates neutral, mixed, ambiguous, and gray do not.

7. Interpretive frame: relation holding before prediction

The depth profile suggests a useful interpretation. In these probes, the relation often remains geometrically coherent across most of transformer depth and only decoheres near the final readout, where the model must resolve the residual state into a next-token distribution. This is consistent with information-bottleneck language: intermediate representations can preserve relational structure while later stages compress that structure toward prediction.

The present study does not explain why the geometry behaves this way. It shows that it does: relation-fields can remain legible in residual space before being compressed or dispersed at the output-facing stage. Relation visibility is not identical to output commitment.

The clean claim is: the residual stream holds the relation; the output-facing stage compresses it toward prediction. That is a descriptive pattern in this probe, not yet a causal mechanism.

8. Findings

Finding 1: Relation-fields become visible across depth

Named poles become geometrically co-situated in the residual stream. This matters, but it is not the main result by itself because raw cosine must be interpreted against anisotropy and unrelated-pair baselines.

Finding 2: Pole co-situation does not imply middle seating

This is the central result. High pole cosine can coexist with failed middle seating. In heads/tails, the poles reach approximately 0.93 cosine under matched/natural carriers, while every candidate middle remains off-axis or prompt-induced. Pole co-situation therefore cannot be treated as evidence of scalar middle structure.

Finding 3: True dichotomies constrain interpolation

The residual stream did not automatically smooth corpus-stable binaries into continua. In heads/tails and even/odd, the model strongly co-situated the poles while still failing to produce stable midpoint candidates. The model can represent a relation-field without inventing a middle.

Finding 4: Span-level analysis is necessary for morally complex oppositions

The good/bad case shows that single-token analysis is insufficient for morally complex oppositions. Phrase-level candidates produced promising midpoint behavior, but because the passing phrases contain both pole tokens, lexical-echo controls are required before treating this as evidence of moral middle structure.

Finding 5: Abstract naming can weaken relation visibility relative to use

The explicit-dichotomy carrier produced lower cosine than natural usage for some pairs. Naming the relation abstractly weakened co-situation relative to using the words in situ. This connects the empirical layer to the theoretical spine of the project: relation-fields may be more strongly instantiated through use than through metalinguistic naming.

Finding 6: The probe classifies named oppositions by middle behavior

The methodological payoff is classification by middle behavior, not by pole proximity alone. The v2 probe distinguishes at least three residual-geometry profiles:

1. Binary relation-field without stable middle: heads/tails and even/odd.
2. Scalar relation-field with stable middle candidates: hot/cold and wet/dry.
3. Span-dependent relation-field requiring phrase-level controls: provisional good/bad.

9. Implications for mechanistic interpretability

First, residual representations can be used to study relation-field structure. The claim is not that the residual representation is meaning. The claim is that the residual representation is a useful unit for observing how corpus-shaped relations become geometrically legible across depth.

Second, MI probes should distinguish relation visibility from field exhaustion. A named opposition can be visible in residual space without being exhausted by two poles. This matters for studying moral, social, legal, scientific, and conceptual oppositions where binary naming may conceal scalar or phrase-level middle structure.

Third, single-token analysis can miss important structure. The good/bad result is not yet a clean moral-middle result, but it is enough to show that span-level representation must be specified and tested.

Fourth, carrier variation should be a first-class experimental factor. The carrier frame is not merely wording. It helps produce the residual state being measured.

10. Limitations and v3 controls

This is a descriptive geometric probe, not a causal mechanistic explanation. It does not show that the model determines truth, morality, or dichotomy. The machine determines nothing; the reader interprets; proximity and seating are corroboration only.

The phrase-level good/bad result is provisional. The phrases neither good nor bad and both good and bad contain both pole tokens. v3 must audit whether phrase candidates were represented by first token, last token, mean-pooled span, or another method. It should add lexical-echo controls that contain both poles but are not middles, and non-pole middle phrases that do not contain good or bad.

Raw cosine_AB needs an anisotropy-aware baseline. v3 should plot unrelated-pair cosine under the same carriers and report cosine_AB_excess.

The 0.65 remainder_vs_control threshold needs sensitivity reporting at 0.55, 0.60, 0.65, and 0.70, especially for phrase-level good/bad candidates near the boundary.

Carrier stability is not meaningful when matched and natural carriers are textually identical. v3 should mark those cases as trivial_identical_carrier or not applicable.

The scalar/dichotomy divergence should not be read as the model recovering logical truth conditions. A conservative interpretation is that residual geometry makes distributional density visible: scalar middle terms such as warm are densely stabilized in relation to both poles, while edge is frame-adjacent to coin but not stabilized as a middle between heads and tails.

11. Conclusion

The contribution of the probe is not that it discovers dichotomies in the model. The contribution is that it separates relation visibility from field exhaustion. A transformer can make a named opposition highly visible in residual space without manufacturing a middle; conversely, scalar relation-fields can produce stable midpoint candidates.

The dichotomy probe shows that the residual stream can hold relation-structure as geometry before final prediction. Relation visibility is not identical to output commitment. This is why the evolving residual representation is the right Y-unit: it is where corpus-shaped relation-structure becomes computationally legible before final readout.

The strongest version of the finding is simple: pole co-situation does not imply middle seating. That dissociation gives a residual-geometric method for classifying named oppositions by middle behavior rather than by pole proximity alone.

Source note for figures

All figures use Gemma 2 2B base with 27 residual states across layers 0 to output. Measurements are descriptive only: the machine determines nothing; the reader interprets; proximity and seating are corroboration only. remainder_vs_control is a candidate's off-axis remainder divided by the pair's unrelated-word baseline; values below 1 indicate the candidate is closer to the pole axis than an unrelated word.

Annotated references

This annotated reference section is organized around the specific claims made by the dichotomy probe: relation visibility, middle seating, late-layer decoherence, and the Y-unit claim that residual representations are the measured process-bearing unit.

A. Core mechanistic interpretability and transformer literature

Elhage et al. (2021). A mathematical framework for transformer circuits.

Use for: residual stream as the central read/write object; attention and MLP components writing into the residual stream. This is the architectural background for treating the evolving residual representation as the measured Y-unit.

Vaswani et al. (2017). Attention is all you need.

Use for: transformer architecture, attention mechanism, residual connections, and feed-forward layers.

Geva et al. (2021). Transformer feed-forward layers are key-value memories.

Use for: MLP/feed-forward layers as content-sensitive, memory-like transformations.

Meng et al. (2022). Locating and editing factual associations in GPT.

Use for: causal tracing and hidden-state/MLP-mediated computations important for factual prediction.

Wang et al. (2023). Interpretability in the wild: A circuit for indirect object identification in GPT-2 small.

Use for: end-to-end circuit explanation, activation patching, and MI evaluation criteria such as faithfulness, completeness, and minimality.

Olsson et al. (2022). In-context learning and induction heads.

Use for: attention heads as identifiable relational mechanisms and in-context relation use.

Belrose et al. (2023). Eliciting latent predictions from transformers with the tuned lens.

Use for: layerwise hidden-state readout and transformers as iterative refinement across depth. Use cautiously because learned translators may smooth late-layer dynamics.

TransformerLensOrg. (n.d.). TransformerLens documentation: Activation patching.

Use for: future-work methodology; activation patching as causal intervention on residual states and other activations.

B. The first methodological threat: representation geometry and anisotropy

Ethayarajh (2019). How contextual are contextualized word representations? Comparing the geometry of BERT, ELMo, and GPT-2 embeddings.

Use for: the foundational demonstration that contextualized representations are anisotropic at every layer. This is the direct source of the objection that raw cosine curves may reflect generic drift rather than relation-specific geometry. The fix is to measure pole co-situation relative to unrelated-pair baselines, structurally parallel to remainder_vs_control.

Gao et al. (2019). Representation degeneration problem in training natural language generation models.

Use for: narrow-cone / representation-degeneration pressure in language generation models, especially near output-facing spaces. This helps bound interpretation of any late-layer or output-facing geometry.

C. Closest precedent for layerwise representation trajectories

Voita, Sennrich, and Titov (2019). The bottom-up evolution of representations in the Transformer: A study with machine translation and language modeling objectives.

Use for: the closest precedent to the depth-trajectory part of the dichotomy probe. They study how token representations and feature-space structure evolve across transformer layers using CCA and mutual-information estimators. Their language-model result is directly relevant: moving from bottom to top layers, information about the past diminishes while predictions about the future are formed. The dichotomy probe begins where this line leaves off: it tracks whether a named relation-field remains geometrically coherent, admits middle seating, or decoheres near the output-facing stage.

D. Prediction formation and final-layer selection

Geva et al. (2022). Transformer feed-forward layers build predictions by promoting concepts in the vocabulary space.

Use for: the view of token representations as changing distributions over the vocabulary, with FFN layers applying additive updates that promote human-interpretable concepts. The late-layer decoherence claim sits inside this account: if late layers promote a single winning concept, relational proximity between competing poles should come under pressure near the output-facing stage.

Geva et al. (2023). Dissecting recall of factual associations in auto-regressive language models.

Use for: a staged account of how information moves to the prediction position across depth, including upper-layer extraction work.

Lioubashevski et al. (2024). Looking beyond the top-1: Transformers determine top tokens in order.

Use for: evidence that computation continues after top-1 prediction saturation, with lower-ranked tokens still being ordered in late layers. This supports treating final layers as active selection, not passive readout.

Wu et al. (2026). Your unembedding matrix is secretly a feature lens for text embeddings.

Use for: an alternative mechanism and limitation. Good/bad decoherence at the output-facing stage could be partly a frequency-geometry or unembedding-space effect rather than purely relational. The present paper need not resolve this, but it should acknowledge it.

E. Relational representation

Hernandez et al. (2024). Linearity of relation decoding in transformer language models.

Use for: the strongest existing case that relations are first-class computational objects in transformer computation. They show that for a subset of relations, subject-to-object mappings are well approximated by linear transformations, while other relations are behaviorally available but not linearly encoded. This is a natural interlocutor: they treat relations as decodable functions; the dichotomy probe treats relations as geometric fields with internal structure: poles, middles, and off-axis remainder.

F. Information-theoretic framing

Tishby and Zaslavsky (2015). Deep learning and the information bottleneck principle.

Use for: theoretical language of compression toward output. The dichotomy probe uses this as suggestive vocabulary only: relation structure may remain visible in intermediate residual states before final output-facing compression.

Shwartz-Ziv and Tishby (2017). Opening the black box of deep neural networks via information.

Use for: empirical companion to the information-bottleneck account. Cite cautiously because compression-phase claims have been contested.

G. Trajectory geometry

Valeriani et al. (2023). The geometry of hidden representations of large transformer models.

Use for: intrinsic-dimension analysis across transformer depth and the idea that residual representations pass through distinct geometric regimes.

Manson (2025). Curved inference: Concern-sensitive geometry in large language model residual streams.

Use for: trajectory-focused geometric interpretability, including curvature and salience vocabulary. The pullback-metric move is relevant to possible objections that raw residual-space distances are coordinate-dependent.

H. Vygotsky / X framework

Vygotsky (1978). Mind in society: The development of higher psychological processes.

Use for: speech/sign mediation, perception and attention, internalization, and cultural development of higher psychological functions. This source supports the X-side unit-of-analysis framing but should not be used to claim that transformers instantiate human word meaning.

Vygotsky (1986). Thought and language.

Use for: word meaning as the unit of analysis, the unity of thought and speech, and verbal thought. In this paper, X motivates the methodological problem; Y is the empirical object measured in transformer residual geometry.

I. Project materials

Chambers Palmer (2026). Token Biography / Dichotomy Transformation Probe v2 [Code, preregistration, measurements, and figures]. GitHub repository.

Use for: present study materials, preregistration links, generated figures, and measurements. Repository: jenniferchamberspalmer-research/token-biography. Figure branch/folder: jenniferchamberspalmer-research/What-if-Conceptual-Turn-in-MI-, branch dichotomy-probe, outputs/dichotomy_probe_v2/figures/.

Chambers Palmer (2026). Dichotomy Transformation Probe - Results v2.

Use for: all reported v2 results, including carrier conditions, cosine_AB, midpoint_t, remainder_vs_control, carrier stability, descriptive classes, and figure source note.

Reading order

1. Ethayarajh (2019) and Gao et al. (2019): these determine whether Figure 1 curves survive review.

2. Voita et al. (2019): the closest precedent to the layerwise trajectory intuition.

3. Geva et al. (2022) and Lioubashevski et al. (2024): late-layer prediction formation and selection.

4. Hernandez et al. (2024): the natural interlocutor on relations as computational objects.

5. Vygotsky (1978, 1986): the X-side unit-of-analysis rationale, kept separate from the measured Y-unit.